although each has a reputation for carrying out research with great precision, their conclusions do not agree. George claims that the circular radius is R = 10 times the Planck length, while Gracie claims that the circular radius is R = 1/10 times the Planck length.

“Gracie,” says George, “based on my string theory calculations, I know that if the circular dimension has radius 10, then I should expect to see strings whose energies are listed in Table 10.1. I have done extensive experiments using the new Planck energy accelerator and they have revealed that this prediction is precisely confirmed. Therefore, with confidence, I claim that the circular dimension has radius R = 10.” Gracie, in defense of her claims, makes exactly the same remarks except for her conclusion that the list of energies in Table 10.2 is found, confirming that the radius is R = 1/10.

In a flash of insight, Gracie shows George that the two tables, although arranged differently, are actually identical. Now George, who, as is well known, reasons a bit more slowly than Gracie, replies, “How can this be? I know that different values for the radius give rise, through basic quantum mechanics and the properties of wound strings, to different possible values for string energies and string charges. If we agree on the latter, then we must agree on the radius.”

Gracie, using her newfound insight into string physics replies, “What you say is almost, but not quite, correct. It is usually true that two different values for the radius give rise to different allowed energies. However, in the special circumstance when the two values for the radius are inversely related to one another—like 10 and 1/10—then the allowed energies and charges are actually identical. You see, what you would call a winding mode 1 would call a vibration mode, and what you would call a vibration mode 1 would call a winding mode. But nature does not care about the language we use. Instead, physics is governed by the properties of the fundamental ingredients—the particle masses (energies) and the force charges they carry. And whether the radius is R or 1/R, the complete list of these properties for the fundamental ingredients in string theory is identical.”

In a moment of bold comprehension, George responds, “I think I understand. Although the detailed description you and I might give for strings may differ—whether they are wound around the circular dimension, or the particulars of their vibrational behavior—the